

OptiCrop: Solution Requirements

Functional Requirements

- The system shall accept 7 numeric soil/climate inputs from the user.
- The system shall return a single recommended crop name based on the inputs.
- The system shall be accessible through a web browser without requiring installation.
- The system shall group crops into seasonal categories for interpretability.

Non-Functional Requirements

- Prediction response time should be near-instant (sub-second) for a single request.
- The model should achieve high classification accuracy on held-out test data.
- The application should run on commodity hardware without GPU requirements.

Out of Scope (Current Version)

- User accounts / login and prediction history storage
- Real-time weather API integration
- Mobile native application